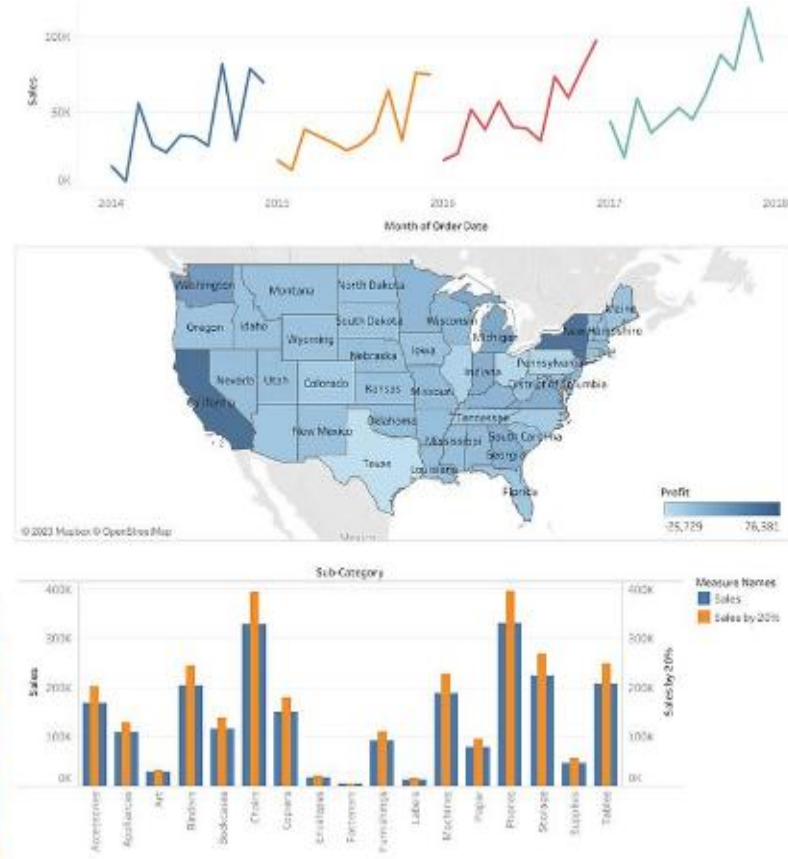
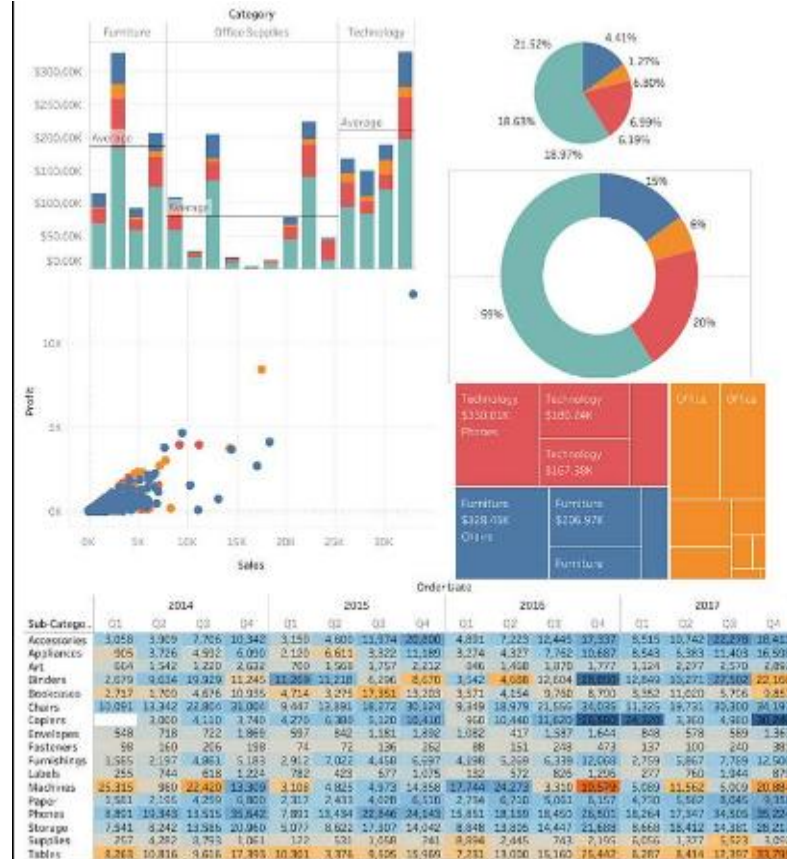


Tableau

Tableau, it is a Business Intelligence (BI) and data visualization tool used to convert raw data into interactive charts, dashboards, and reports.



Key features of Tableau

Feature	Description
Data Visualization	Creates graphs, charts, maps, and plots
Dashboard	Combines multiple visualizations into one interactive screen
Data Connection	Connects to Excel, CSV, SQL databases, cloud data, etc.
Drag and Drop	Most visualizations can be created without coding
Interactive Analysis	Filters, drill-downs, and interactive charts
Data Blending	Combines data from different sources
Real-time Analysis	Can work with live database connections
Sharing	Dashboards can be published and shared with others

Tableau Software is an American interactive data visualization software company focused on business intelligence. It was founded in 2003 in Mountain View, California and is headquartered in Seattle, Washington.

Founders	<u>Pat Hanrahan, Christian Chabot,</u> <u>Chris Stolte</u>
Founded	<u>January 2003, Mountain View,</u> <u>California, United States</u>
Headquarters	<u>Seattle, Washington, United States</u>

Main Tableau products

- 1. Tableau Desktop** – Used to create visualizations and dashboards.
- 2. Tableau Cloud** – Cloud-based platform for publishing and sharing dashboards.
- 3. Tableau Server** – Tableau platform deployed within an organization's infrastructure.
- 4. Tableau Public** – Free platform for creating and publicly sharing visualizations.
- 5. Tableau Prep** – Used for cleaning and preparing data.

Experiment -1

Exploring and Visualizing Sales Data. Import a sample sales dataset into Tableau. Create a dashboard that includes the following visualizations:

- a) A bar chart showing total sales by product category.
- b) A line chart showing monthly sales trends over the past year.
- c) A geographic map showing sales distribution by region.
- d) Use a dual-axis chart or side-by-side bar charts to compare sales and return rates across regions.

Experiment-1 Solution: a) **A bar chart showing total sales by product category.**

Step 1: Open Tableau

1. Open Tableau Desktop or Tableau Public.
2. Click Connect to Data.
3. Select Microsoft Excel if your sales dataset is in Excel.
4. Browse and select your sales dataset.
5. Drag the required sheet/table into the data area.

Step 2: Open a worksheet

1. Click Sheet 1 at the bottom.
2. You will see the Data pane on the left.
3. Identify:
 - **Category** → Dimension
 - **Sales** → Measure

Step 3: Place Category on Columns

1. Drag Category from the Data pane.
2. Drop it onto the Columns shelf.

Step 4: Place Sales on Rows

1. Drag Sales from the Data pane.
2. Drop it onto the Rows shelf.
3. Tableau automatically calculates SUM(Sales).



The resulting visualization will be a bar chart showing total sales for each category.

b) A line chart showing monthly sales trends over the past year.

1. Verify the fields

In the Data pane on the left:

- **Order Date** should have a calendar/date icon.
- **Sales** should be a numeric measure.
- Tableau normally identifies **Sales** as SUM(Sales) automatically.

If **Order Date** is not recognized as a date:

Right-click Order Date → Change Data Type → Date.

2. Create the monthly sales trend

Drag:

Order Date → Columns

Then:

Sales → Rows

Initially, Tableau may display sales by year, quarter, or another date level.

3. Change Order Date to Month

Click the drop-down arrow on the **Order Date** pill on the Columns shelf.

Select:

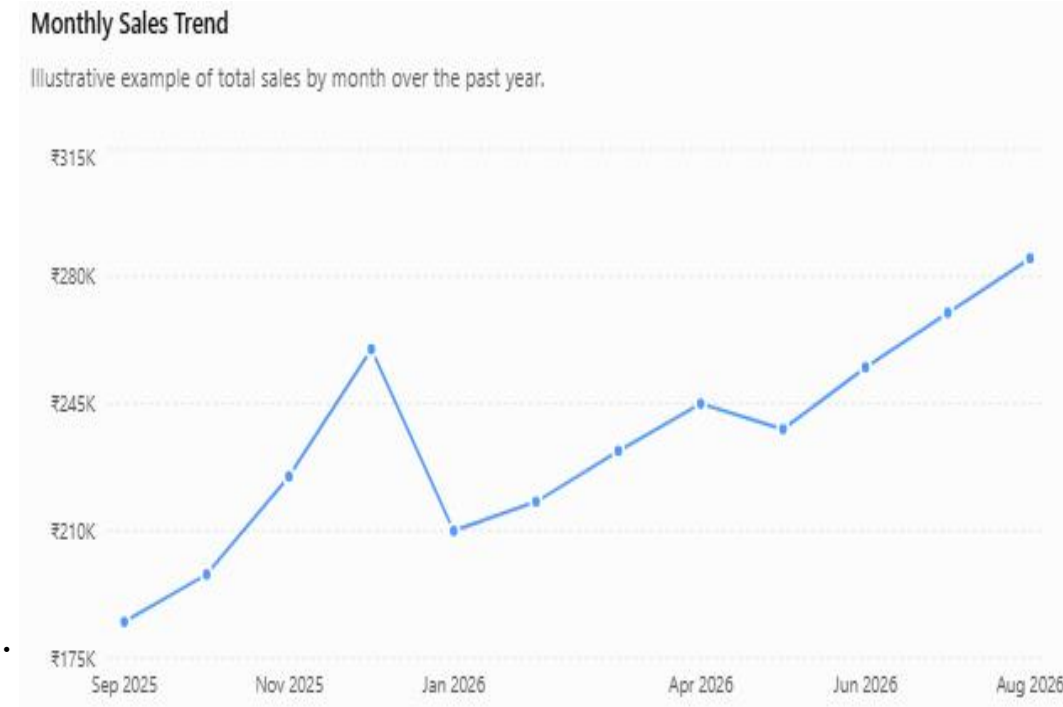
Month

Prefer the continuous Month option if you want a continuous time-series line.

The view should now contain:

Jan → Feb → Mar → ... → Dec

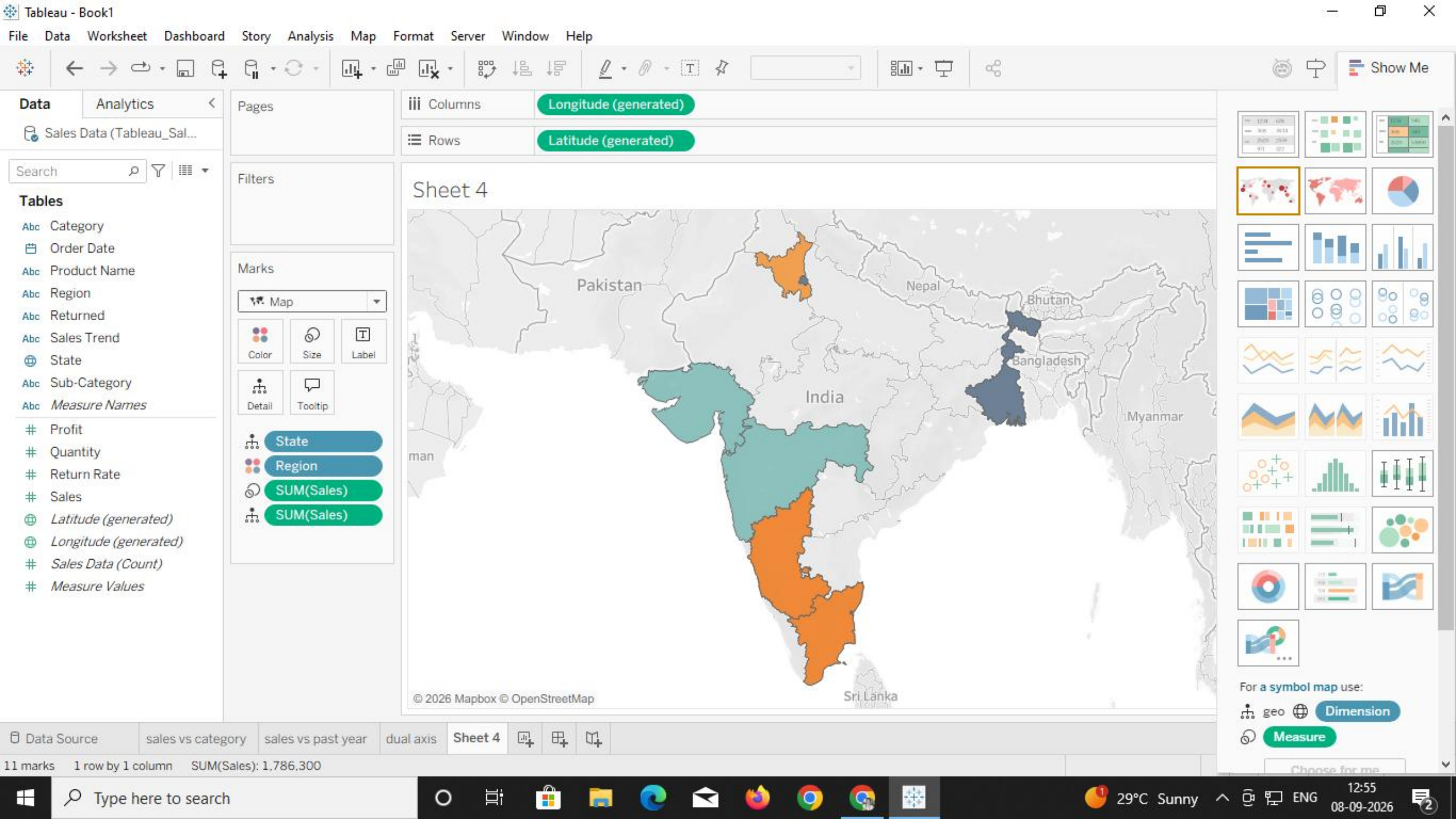
with total sales plotted for each month.



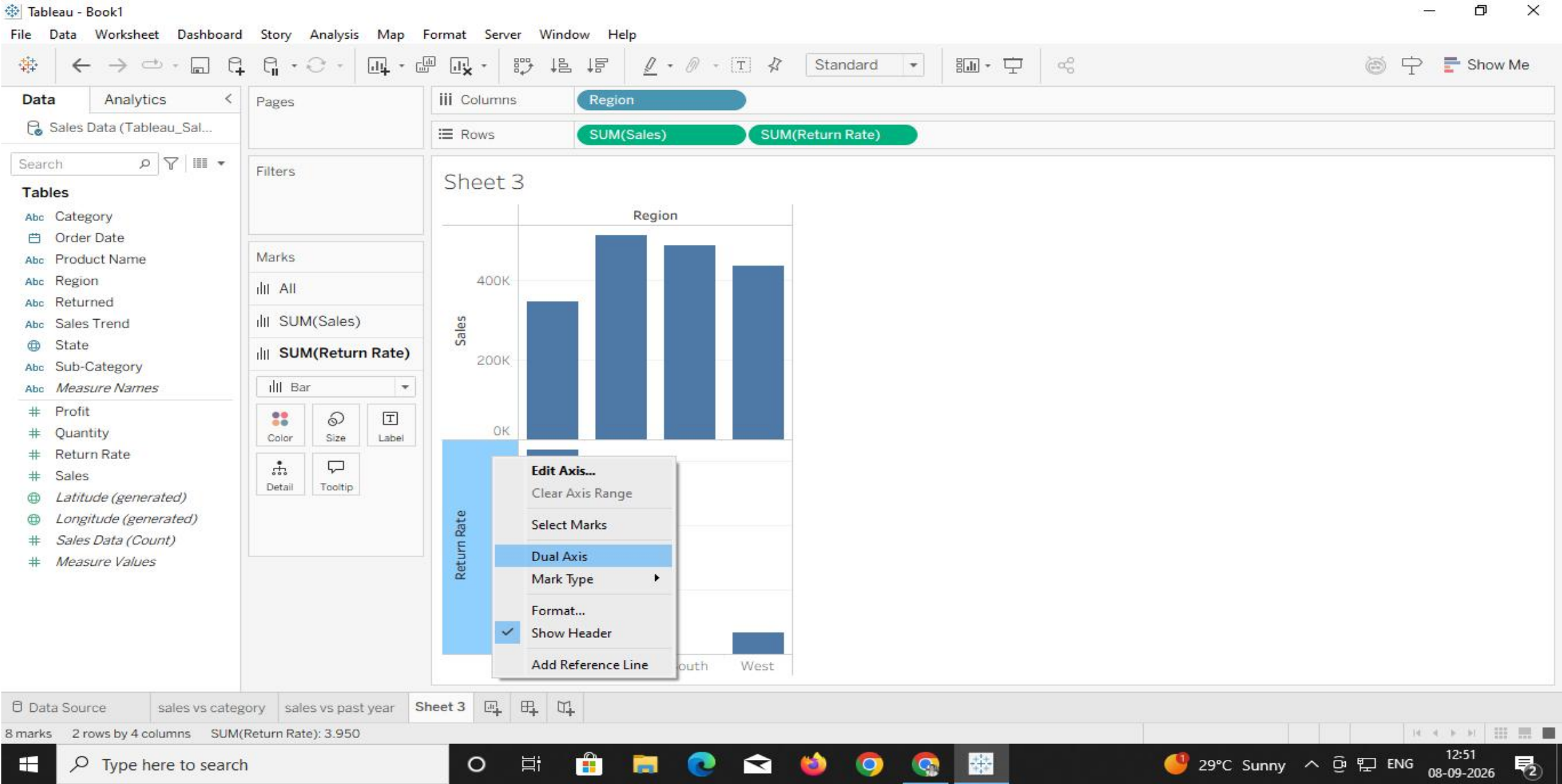
4. Change the chart type to Line

On the Marks card:

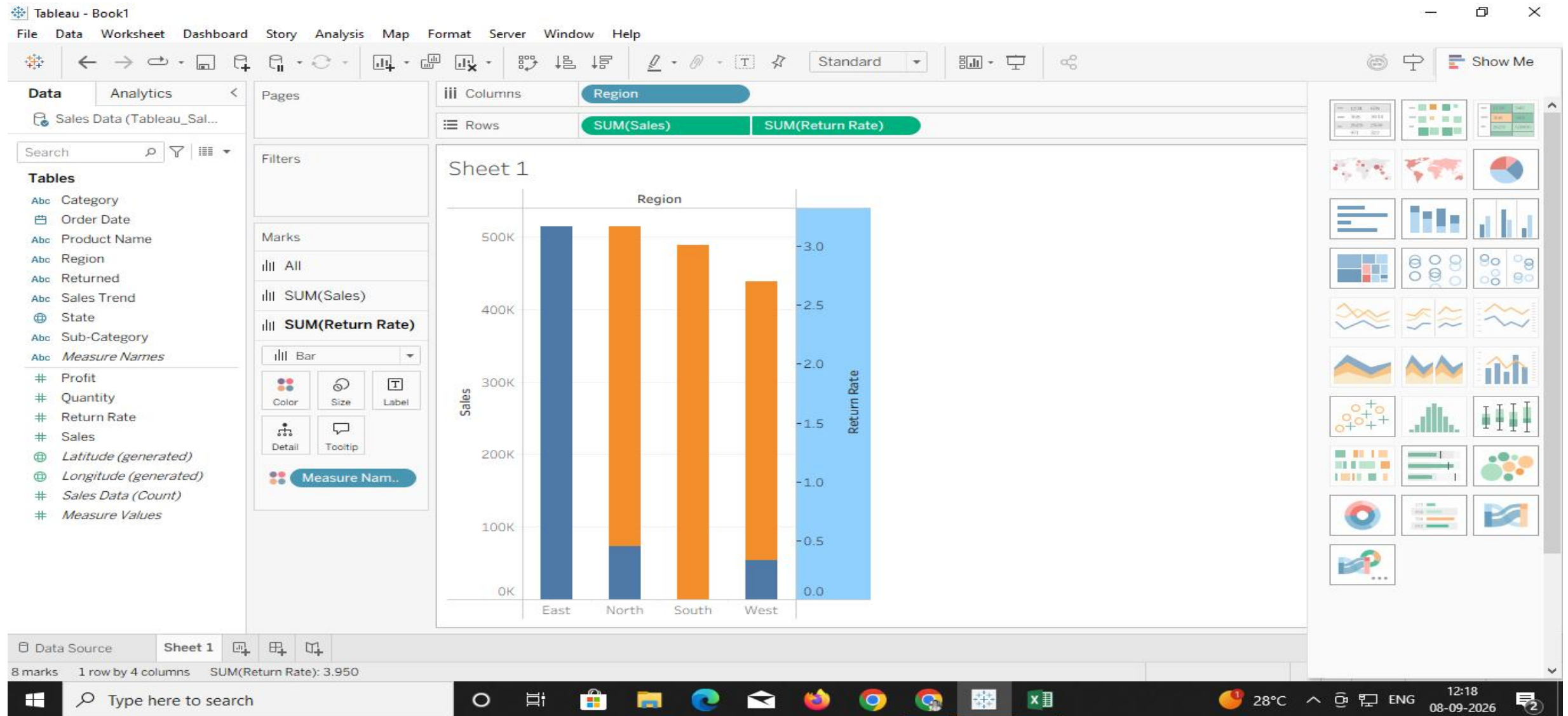
1. Click the chart-type drop-down.
2. Select Line.



d) Use a dual-axis chart or side-by-side bar charts to compare sales and return rates across regions.



Right click on the graph and select dual axis option. Both the graph can be visualized in dual axis



5. To show only the past 12 months:

1. Drag **Order Date** to the Filters shelf.
2. Select Relative Date.
3. Choose Months.
4. Select Last → 12 → Months.
5. Click Apply → OK.

6. Add data labels

To display the sales value for each month:

1. Click Label on the Marks card.
2. Check Show mark labels.

The sales amount will appear near each point on the line.

7. Format the sales values

If the values are large:

1. Right-click the **Sales** axis.
2. Select Format.
3. Under Numbers, select an appropriate number format.
4. You can use currency formatting if your sales data represents monetary values.

For example:

₹185,000 → ₹185K

8. Add a meaningful title

Double-click the chart title and change it to:

Monthly Sales Trends – Past 12 Months

A good chart should clearly communicate:

- X-axis: Month
- Y-axis: Total Sales
- Line: Monthly sales movement

9. Tooltip

Click Tooltip on the Marks card and include:

- Month
- Total Sales
- Category
- Region

For example:

Month: June 2026

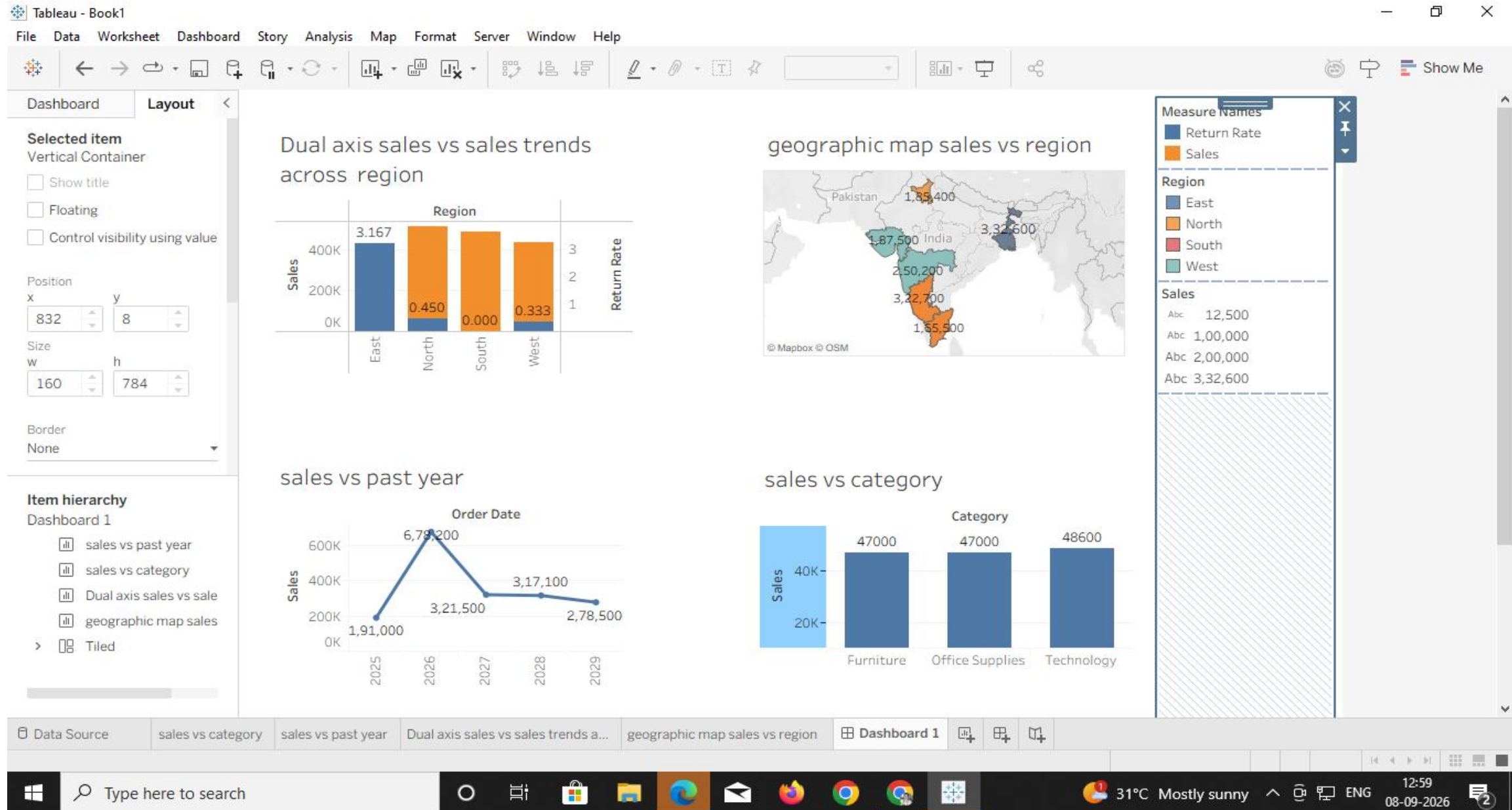
Total Sales: ₹255,000

You can also add a trend line:

Analytics → Trend Line → Linear

This helps identify whether sales are generally increasing or decreasing.

Dashboard



Experiment -2

Using a dataset on customer purchases, create a dashboard with the following advanced visualizations:

- a) A scatter plot showing the relationship between customer age and total spending.
- b) A heat map illustrating product sales intensity by store location.
- c) A dual-axis chart comparing sales and profit margins over time.
- d) Create a bar chart or treemap showing total sales revenue by product and region.

a) A scatter plot showing the relationship between customer age and total spending.

Customer Age vs Total Spending — Scatter Plot

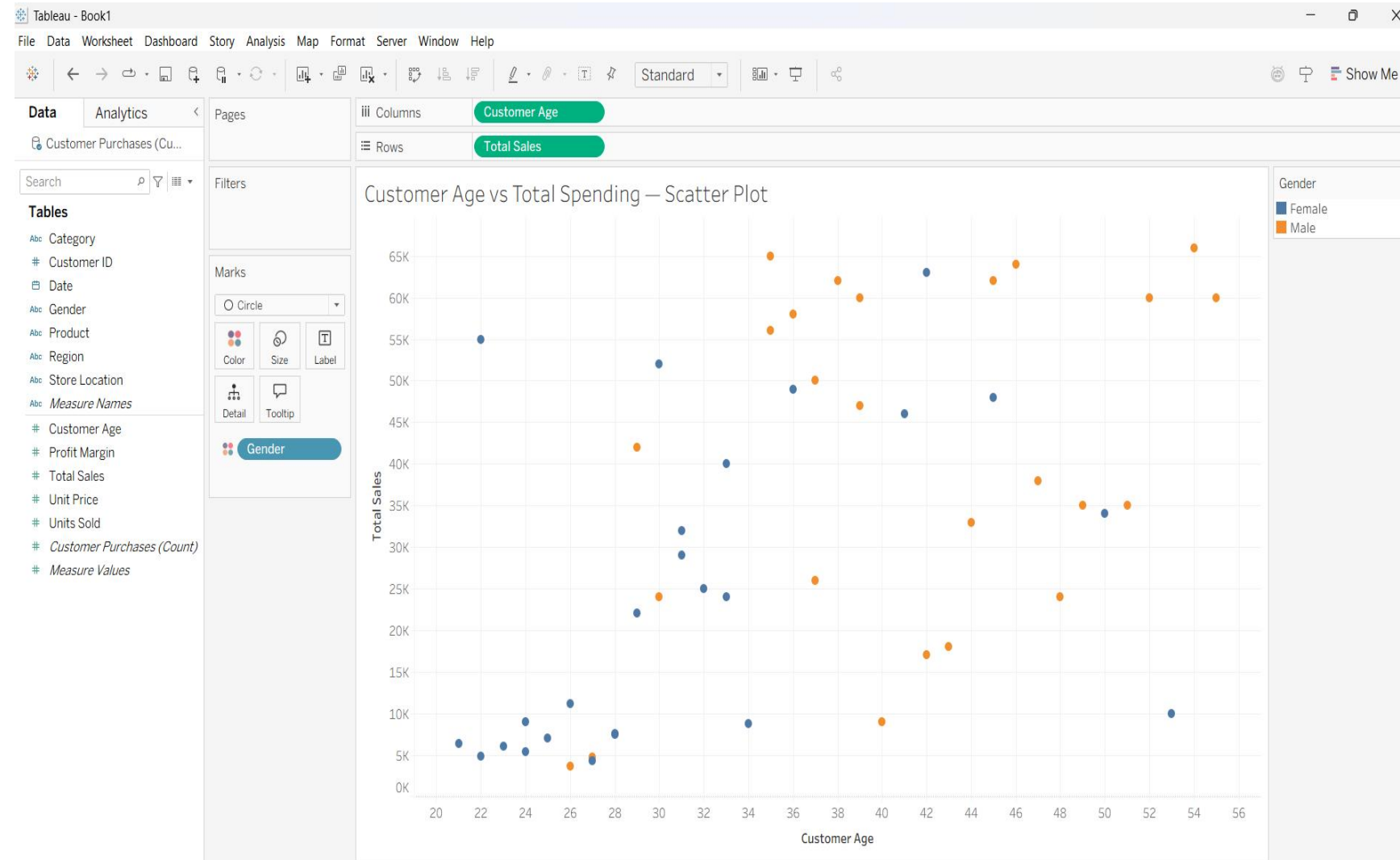
Columns: Customer_Age

Rows: Total_Sales

Marks: Circle

Details: Customer_ID

Color: Gender



b) A heat map illustrating product sales intensity by store location.

Product Sales Intensity by Store Location — Heat Map

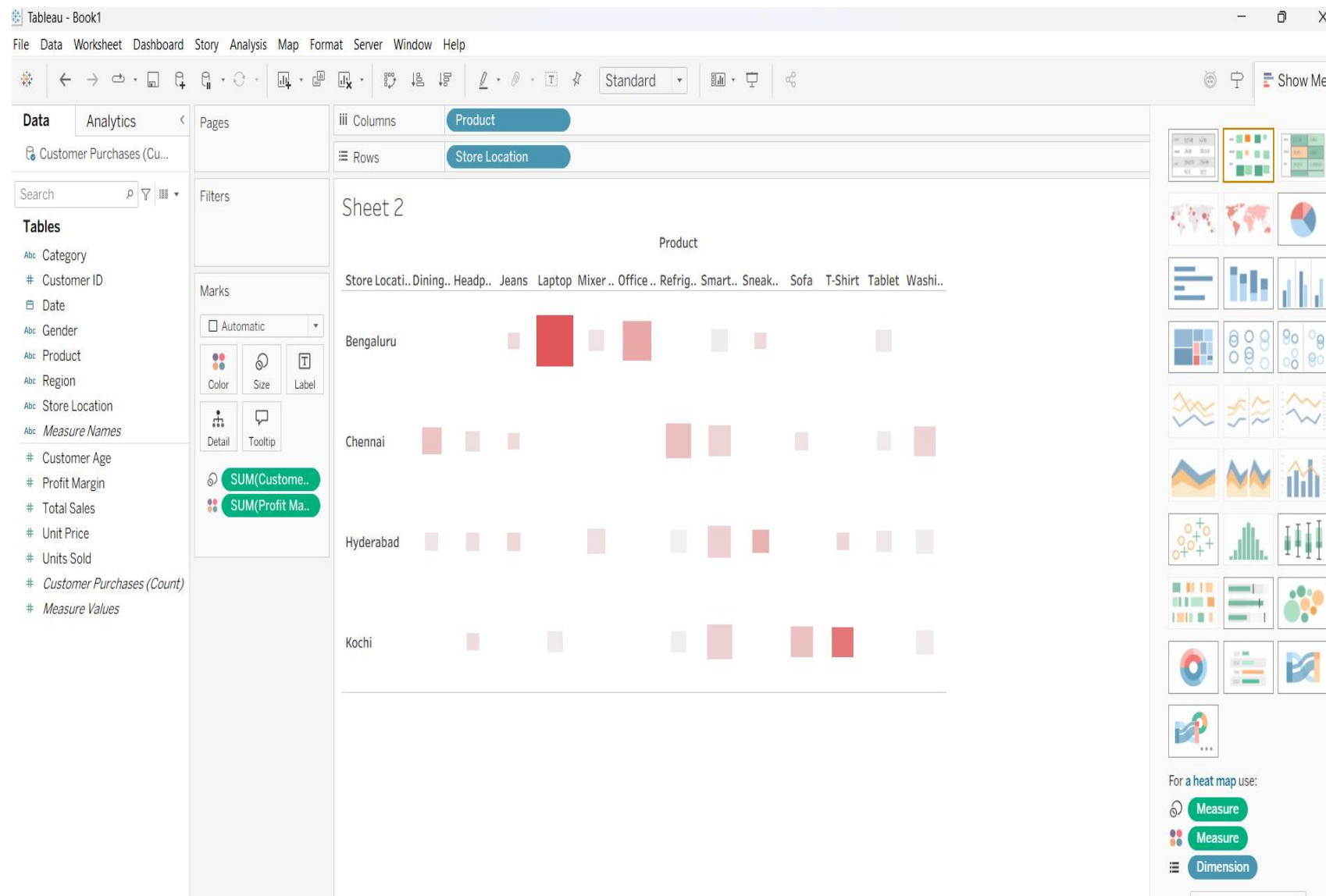
Rows: Store_Location

Columns: Product

Color: SUM(Total_Sales)

Label: SUM(Total_Sales)

Marks: Square



c) A dual-axis chart comparing sales and profit margins over time.

Sales vs Profit Margin Over Time — Dual-Axis Chart

Columns: MONTH(Date)

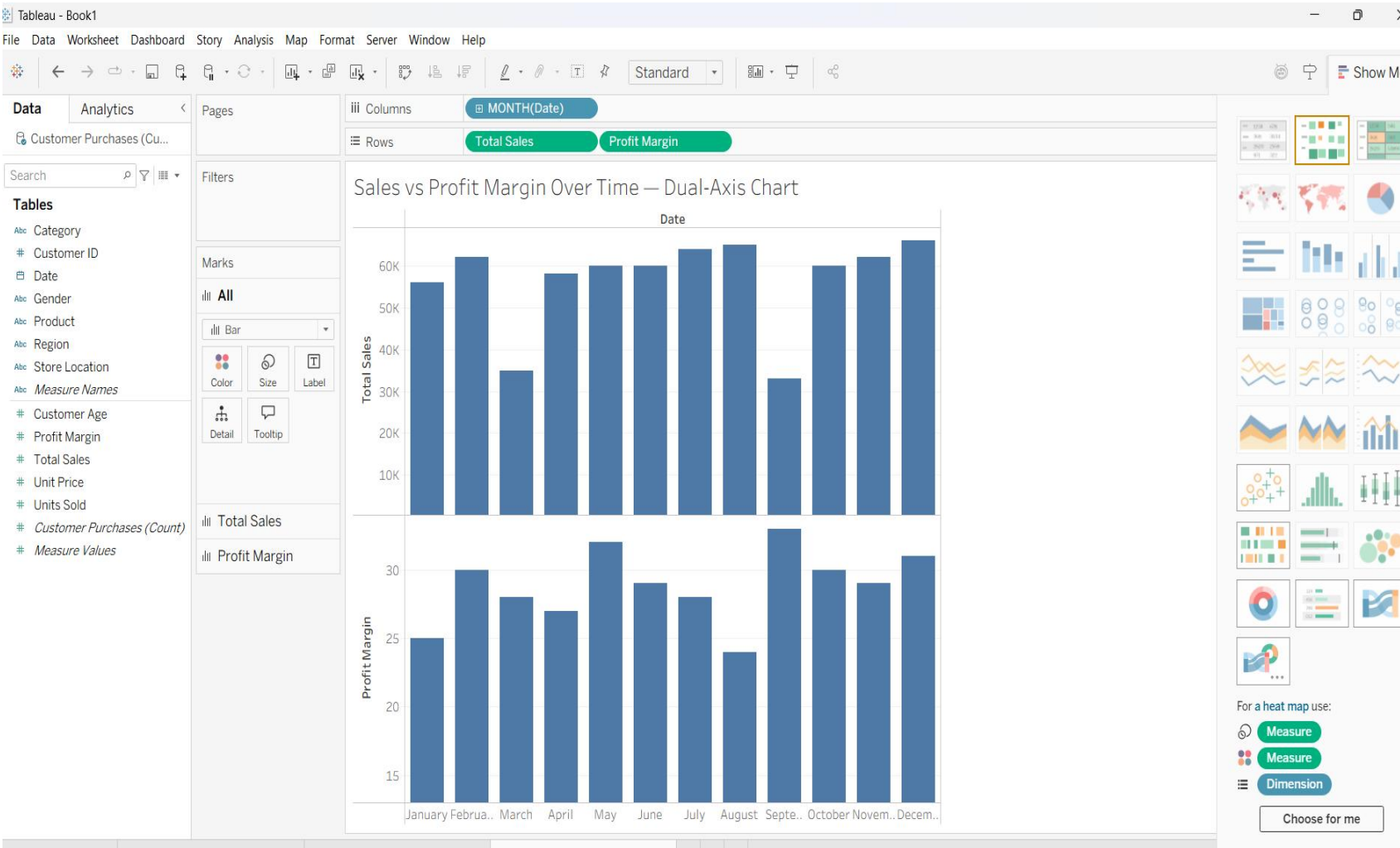
First measure:

Rows: SUM(Total_Sales)

Second measure:

Drag AVG(Profit_Margin) onto the same axis and select:

Right Click on bottom graph select dual axis. It will merge both the graphs



d) Create a bar chart or treemap showing total sales revenue by product and region.

Marks → Treemap

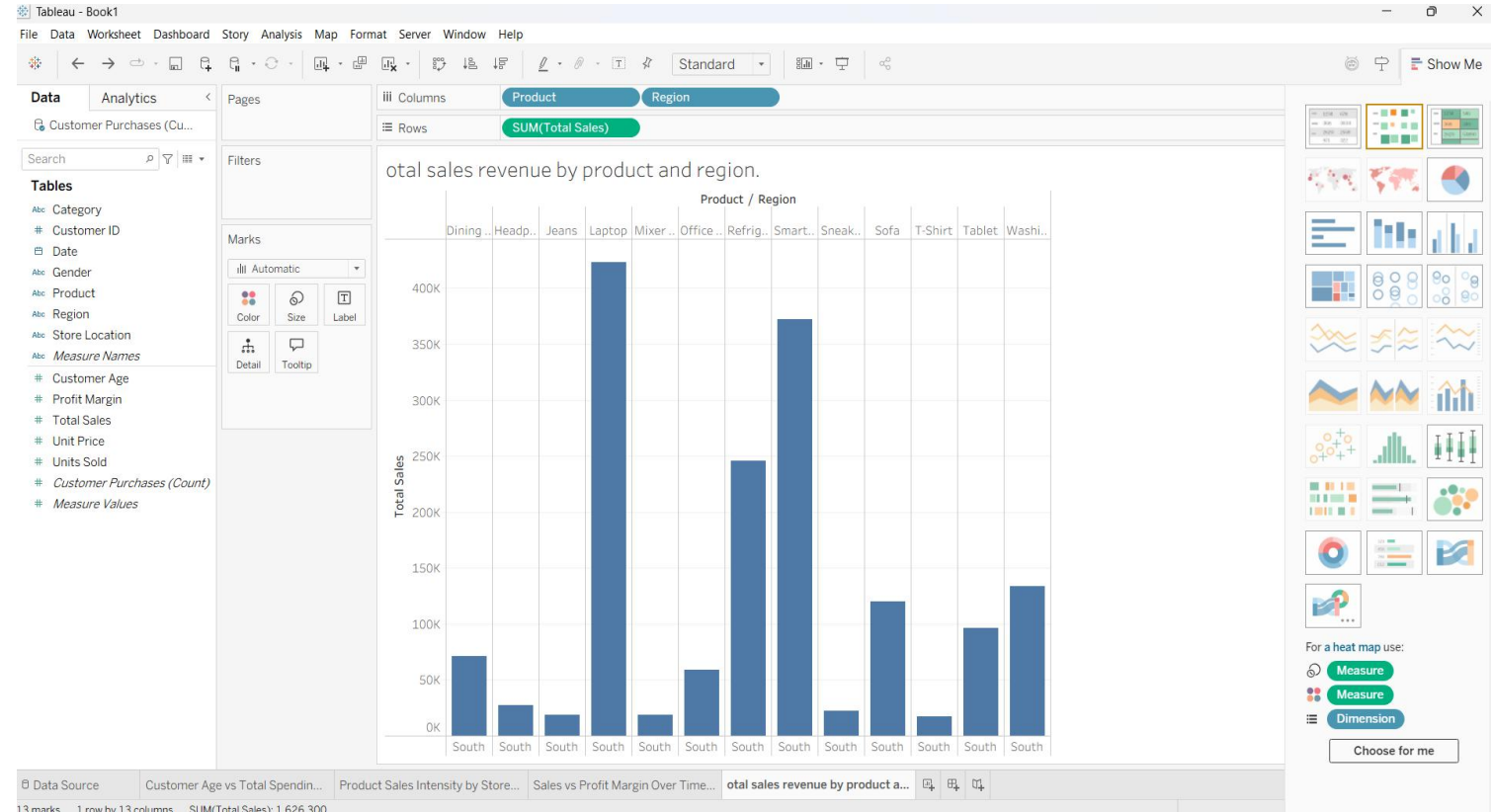
- **Product** → Label
- **Region** → Detail
- **SUM(Total_Sales)** → Size
- **SUM(Total_Sales)** → Color

Alternatively, create a horizontal bar chart using:

Rows: Product

Columns: SUM(Total_Sales)

Color: Region



d) Treemap

- Drag Category to the Label or Marks card.
- Drag Sub-Category to Label.
- Drag Sales to Size.
- Drag Profit to Color.
- Select the required fields.
- Click Show Me → Treemap.

OR

- Drag Category to Detail.
- Drag Sub-Category to Label.
- Drag Sales to Size.
- Drag Profit to Color.
- Change the Marks type to Treemap.

Treemap

Tables

Abc Category1

📅 Order Date

Abc Product Name

Abc Region

Abc Returned

🌐 State

Abc Sub-Category

Abc Measure Names

Profit

Quantity

Return Rate

Sales

🌐 Latitude (generated)

🌐 Longitude (generated)

Sales Data (Count)

Measure Values

Marks

☐ Automatic



Color



Size



Label



Detail



Tooltip



SUM(Profit)



SUM(Sales)

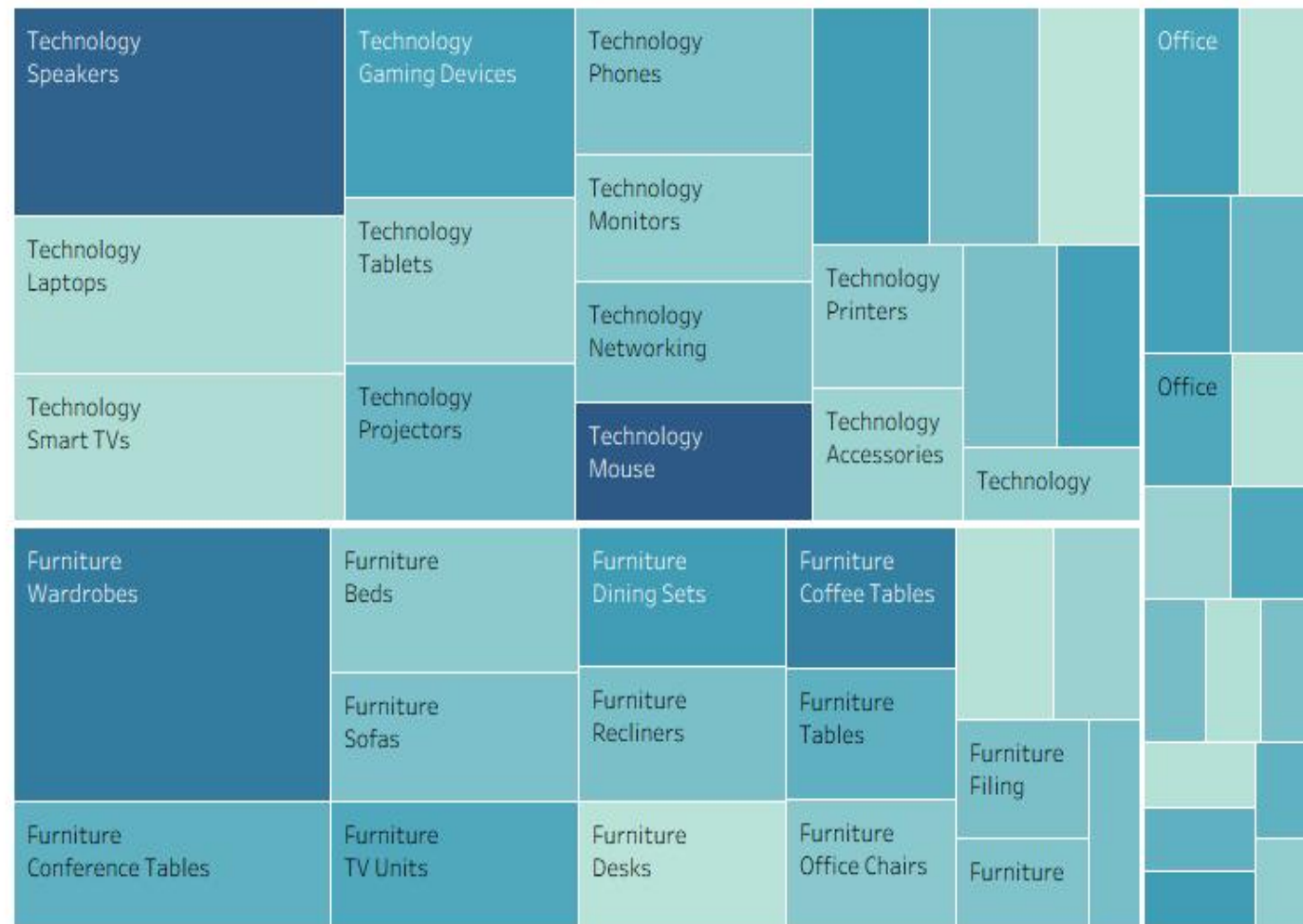


Category1



Sub-Category

SHEET 1



For a treemap use:

Measure

Measure

Dimension

Data Source

Sheet 1

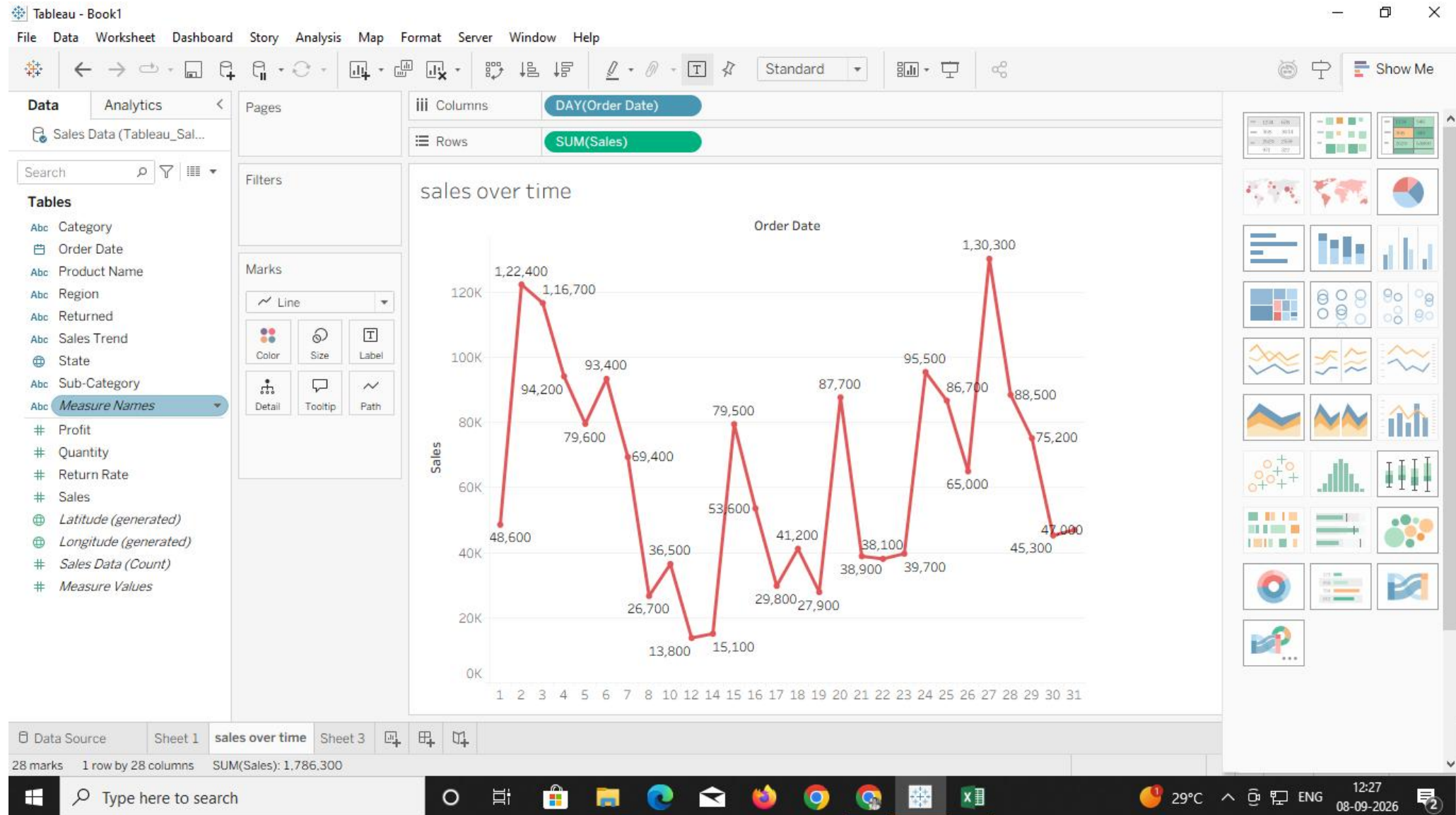


Experiment -3

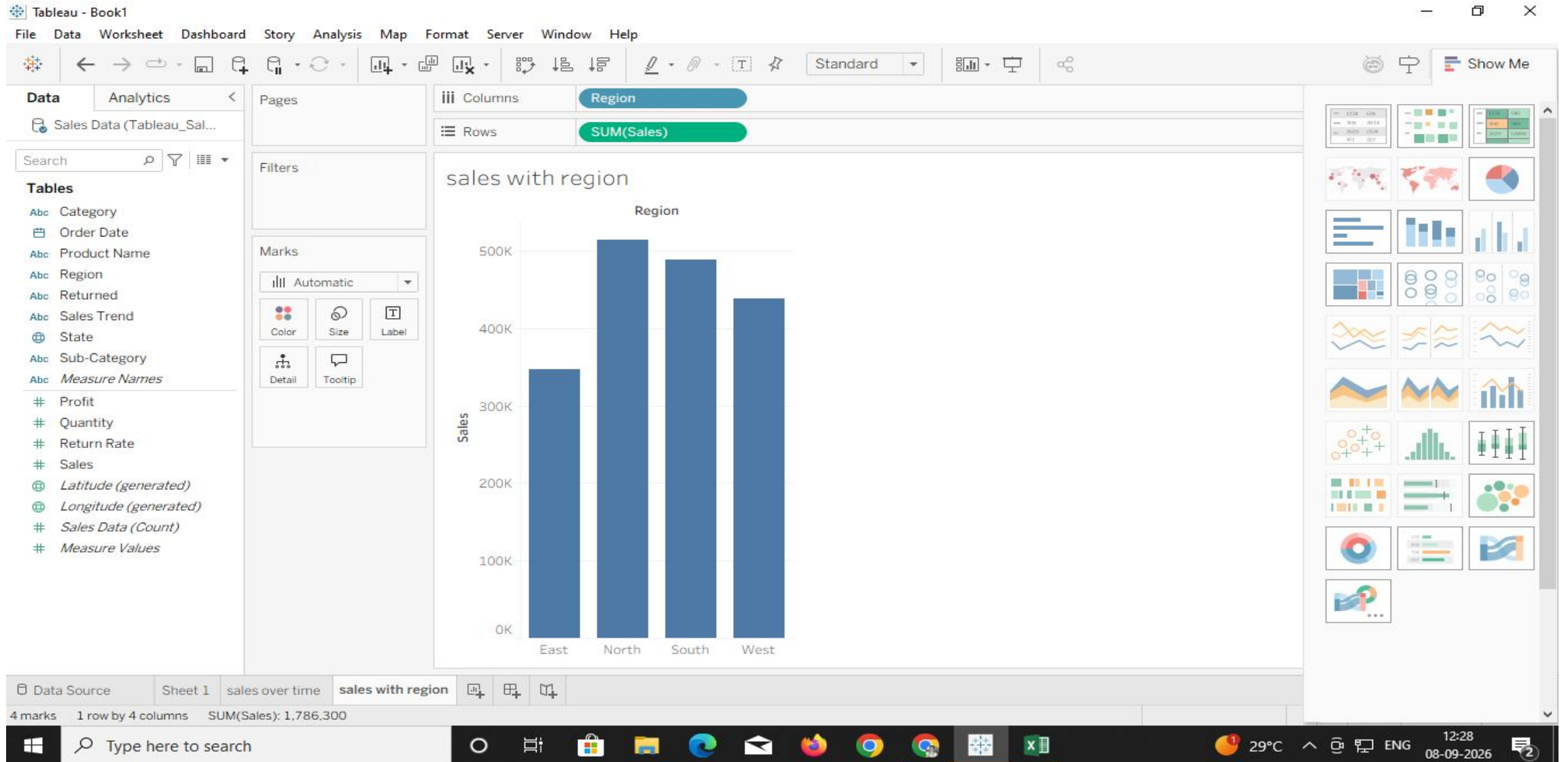
3. Analysing Sales Performance: Create various visualizations to analyze the sales performance.

- a. Create a line chart to show sales trends over time.
- b. Create a bar chart to compare sales across different regions.
- c. Use a pie chart to display the distribution of sales among different product categories.
- d. Create a dashboard combining these visualizations to provide an overall view of sales performance.

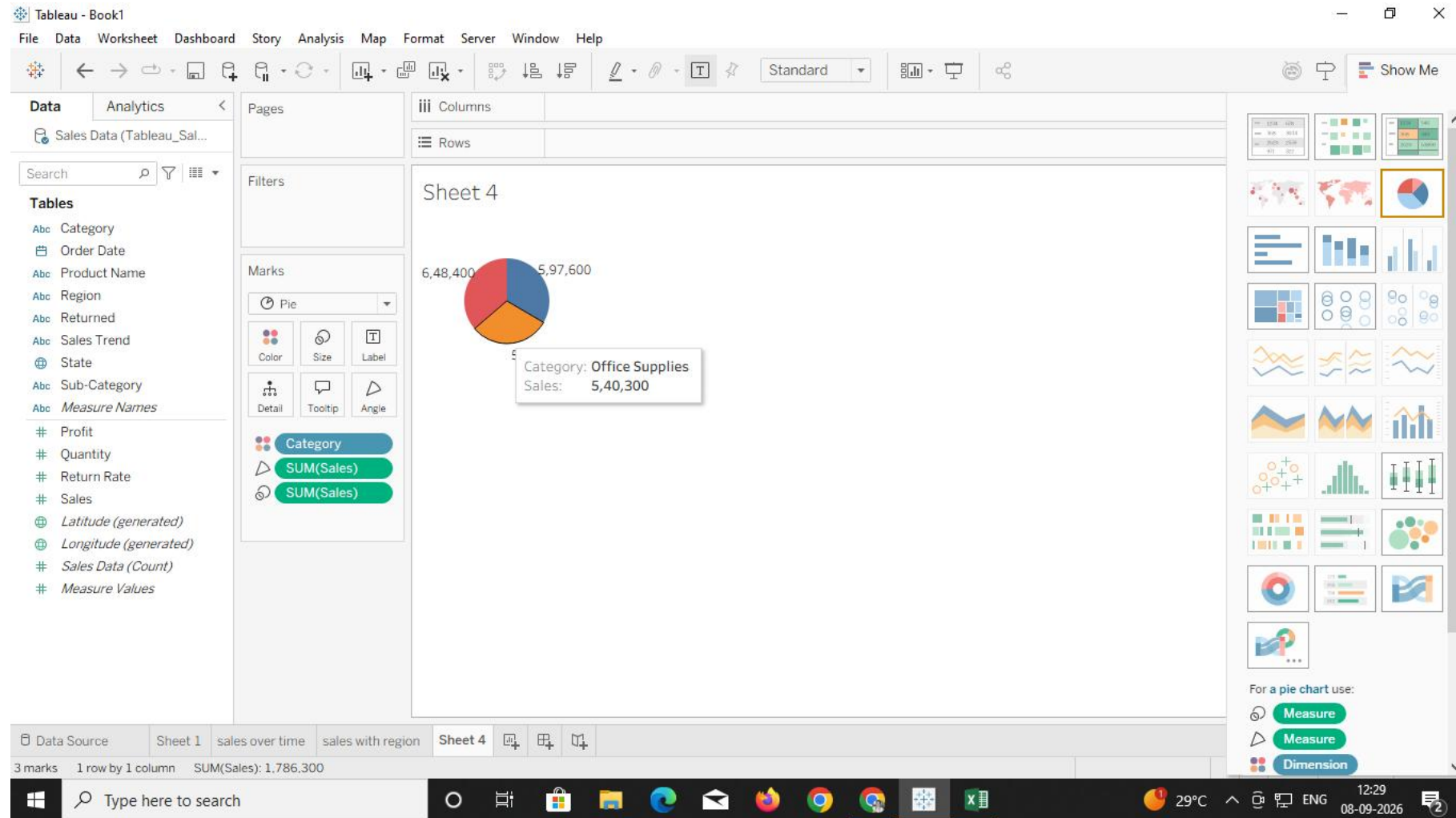
a. Create a line chart to show sales trends over time.



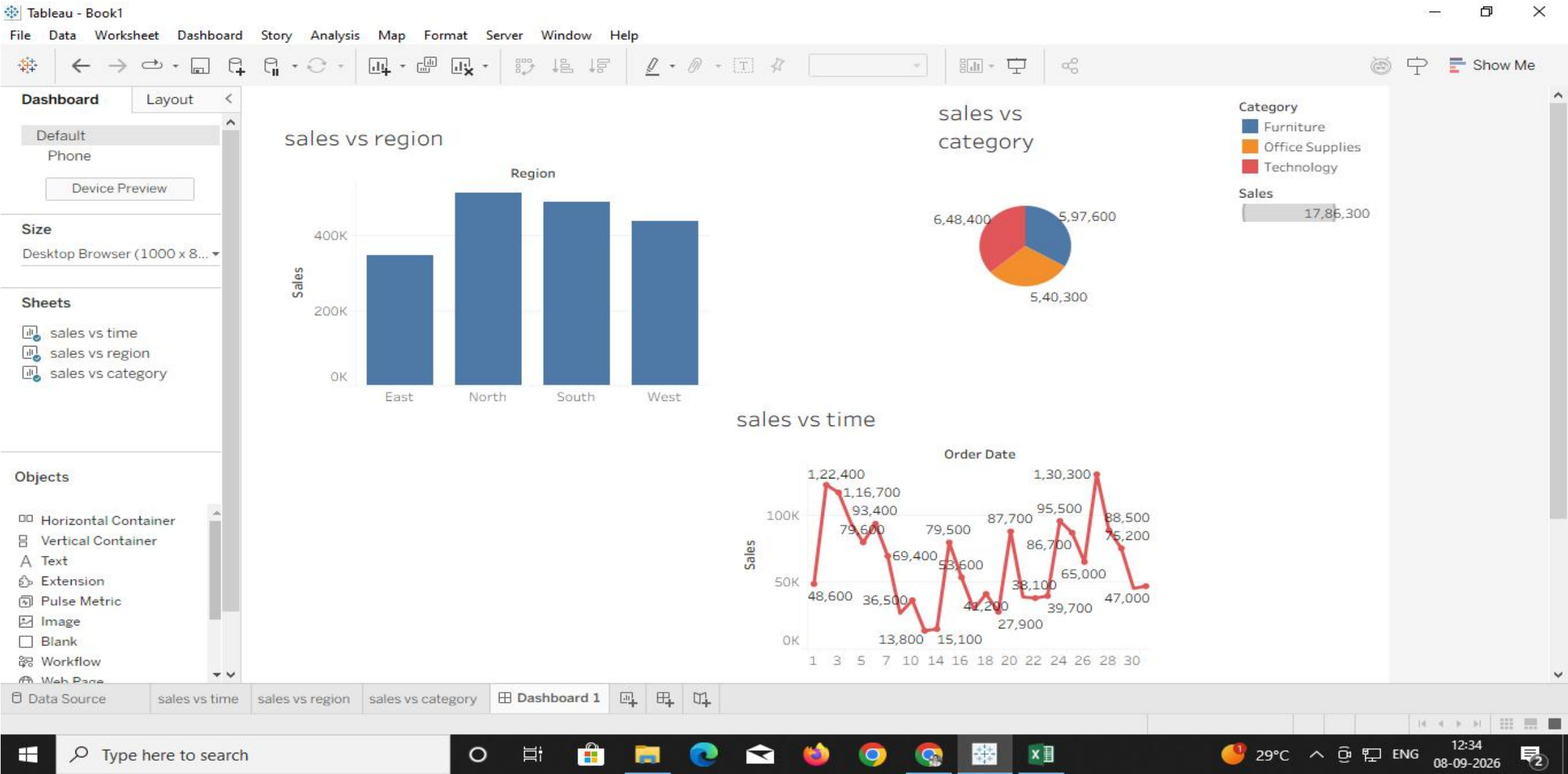
b. Create a bar chart to compare sales across different regions.



C. Use a pie chart to display the distribution of sales among different product categories.



D. Create a dashboard combining these visualizations to provide an overall view of sales performance.



END